A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Fossil fuels & nuclear power: sustainability and new technologies - Set 1

Mark scheme

AQA spec 3.3.3, 3.3.4.1; Energy resources; textbook pp. 183-196

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**1 mark each** 1 mark for each of the 2 points below. **1** Natural pressure of gas above the oil (or dissolved in it) or water beneath forces oil up the production well – pump-jacks ('nodding donkeys') increase flow **1** Pressure falls as oil is removed so the remaining oil no longer flows to the well **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
01.2	**1 mark each** 1 mark for each of the 2 points below. **1** Water or natural gas is pumped down an injection well to maintain the pressure and flow of oil (recovery to about 40%) **1** Captured CO2 can be pumped down instead, increasing oil recovery while storing the CO2 underground **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **carbon capture** → capturing carbon dioxide from flue gases and storing it underground	2

Do not write outside the box

Question	Marking guidance	Marks
01.3	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Steam injection (from combustion or solar parabolic concentrators) to heat the oil and reduce its viscosity **1** Controlled underground combustion to heat the oil **1** Detergents / solvents to reduce surface tension so oil flows more easily **1** Bacteria that partially digest heavy oil into lighter oils and produce CO2 to maintain pressure **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
01.4	**1 mark each** 1 mark for each of the 3 points below. **1** Many wells can be drilled from a single platform, reducing land take / number of rigs **1** Deposits under locations where rigs cannot be placed (urban areas, sensitive habitats) can be reached **1** Drilling can follow softer strata (quicker) and target multiple small reservoirs up to 10 km away, increasing total recovery **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 187-188; AQA spec 3.3.4.1.	
02.1	**1 mark each** 1 mark for each of the 3 points below. **1** Shale has very low permeability so the 'tight' oil / gas cannot flow through the rock to a well **1** Fluid is pumped down at high pressure to open fissures in the shale along which oil / gas can flow to the recovery well **1** Sand grains (to hold fissures open) and solvents are pumped in to increase the recovery rate **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **hydraulic fracturing** → pumping fluid into rock at high pressure to release gas or oil • **permeability** → how easily water passes through / lets water through	3

Do not write outside the box

Question	Marking guidance	Marks
02.2	**1 mark each** 1 mark for each of the 5 points below. **1** Natural gas may enter aquifer water **1** Injected chemicals may reach aquifers or the surface – toxic metals in the rock may be mobilised **1** Large volumes of water needed **1** Earth tremors as fracking releases natural tensions in the crust (though not tremors that could not occur naturally) **1 Reduced by:** collection, treatment and re-use of waste water – well casing standards – restricting sites away from aquifers / sensitive areas – monitoring **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **fracking** → pumping fluid into rock at high pressure to release gas or oil • **aquifer** → a permeable rock that stores groundwater / underground water store	5
02.3	**1 mark each** 1 mark for each of the 2 points below. **1** Extraction is expensive and energy-intensive (quarrying and hot water treatment, steam / solvent / combustion in-situ; heating shale to release oil) so it is not economically viable at current prices **1** Reserves only include what is economically exploitable now – high environmental impacts (land take, water, CO_2) also limit exploitation **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 184, 189-190; AQA spec 3.3.4.1.	
03.1	**1 mark each** 1 mark for each of the 4 points below. **1 Gasification:** coal (including deposits too deep to mine) is burnt underground under controlled conditions to produce fuel gases (hydrogen, carbon monoxide, methane) **1** Allows deep coal to be exploited and gas can be piped / used in gas turbines or for hydrogen with CCS **1 Liquefaction:** coal is converted to liquid hydrocarbons directly with solvents or via gasification and chemical conversion **1** Provides liquid fuels for vehicles / aircraft that solid coal cannot supply **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **CCS** → capturing carbon dioxide from flue gases and storing it underground	4

Do not write outside the box

Question	Marking guidance	Marks
03.2	**1 mark each** 1 mark for each of the 2 points below. **1** A solid, ice-like crystalline structure in which methane is trapped within water ice **1** Found at low temperatures (polar regions / permafrost) or high pressure in ocean sediments around continental margins **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
03.3	**1 mark each** 1 mark for each of the 4 points below. **1 Water heating:** hot water pumped into sediments melts the hydrate releasing methane **1 Depressurisation:** drilling reduces pressure so methane dissociates from the crystals **1 CO2 injection:** CO2 bonds more strongly to the ice and displaces methane (could double as CCS) **1 Risks:** methane not collected escapes to the atmosphere - powerful greenhouse gas / positive feedback – seabed ecology poorly understood – sediment instability / landslides **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **CCS** → capturing carbon dioxide from flue gases and storing it underground • **positive feedback** → a change that makes itself larger / a self-reinforcing cycle	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 187, 190; AQA spec 3.3.4.1.	
04.1	**1 mark each** 1 mark for each of the 3 points below. **1** Nuclei of large atoms (uranium-235, plutonium-239) split when hit by a neutron **1** A small amount of mass is converted to a large amount of energy ($E = mc^2$) and more neutrons are released, causing a chain reaction (controlled by control rods / moderator) **1** The heat boils water to steam which drives turbines and generators **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **control rods** → absorb neutrons to control the rate of fission • **moderator** → slows the neutrons so that fission continues **Do not credit** • 'nuclear power releases no carbon dioxide' - mining, fuel processing, construction and decommissioning all do	3

Do not write outside the box

Question	Marking guidance	Marks
04.2	**1 mark each** 1 mark for each of the 2 points below. **1** Output changes slowly (days) so it cannot follow rapid fluctuations in demand **1** Very high construction cost but low fuel cost means it is most economic running continuously at full output **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'nuclear power releases no carbon dioxide' - mining, fuel processing, construction and decommissioning all do	2
04.3	**1 mark each** 1 mark for each of the 2 points below. **1** Only 30 t of fuel a year (very high energy density) so no need for railways / ports to deliver millions of tonnes of coal **1** Still need condenser cooling water (coast / large lake) and a stable site **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'nuclear power releases no carbon dioxide' - mining, fuel processing, construction and decommissioning all do	2
04.4	1 mark for the point below. **1** Uranium must be mined, purified, concentrated and processed into fuel using (fossil) energy – reactor construction and decommissioning have embodied energy **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **decommissioning** → safely dismantling a power station at the end of its life • **embodied energy** → all the energy used to make and transport a product **Do not credit** • 'nuclear power releases no carbon dioxide' - mining, fuel processing, construction and decommissioning all do	1
04.5	**Any 2 of 4** 1 mark each for any 2 of the 4 points below, in any order, maximum 2. **1** No commercial reactor has been fully decommissioned so costs are unknown and often exceed estimates **1** Large projects often overrun original cost estimates **1** Long-term waste disposal costs uncertain **1** State subsidies may hide costs **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • 'nuclear power releases no carbon dioxide' - mining, fuel processing, construction and decommissioning all do	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 191-193; AQA spec 3.3.4.1.	

Do not write outside the box

Question	Marking guidance	Marks
05.1	**Any 3 of 5** 1 mark each for any 3 of the 5 points below, in any order, maximum 3. **1** Public opposition following accidents (Three Mile Island 1979, Chernobyl 1986, Fukushima 2011) **1** Very complex, expensive technology needing an advanced industrial infrastructure **1** Concerns about links between civil nuclear materials and weapons / terrorism **1** Uncertainty over permanent disposal of radioactive waste **1** Uncertain decommissioning costs **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **decommissioning** → safely dismantling a power station at the end of its life **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • 'nuclear power releases no carbon dioxide' - mining, fuel processing, construction and decommissioning all do	3
05.2	**1 mark each** 1 mark for each of the 3 points below. **1** Only 0.7% of natural uranium is fissile U-235 – the remaining 99.3% is non-fissile U-238 (a 'fertile' fuel) **1** Neutron bombardment in the reactor converts U-238 to fissile plutonium-239 (via beta decays) **1** So much more energy is harnessed from the mined uranium – fast reactors need no moderator but are complex and costly **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **moderator** → slows the neutrons so that fission continues	3
05.3	**2 marks per point** 2 marks for each of the 2 points below; award 1 mark where a point is only partly made. **2 Advantages:** thorium is three times more abundant – much less radioactive waste with shorter half-lives – harder to make weapons material – no fuel enrichment needed **2 Disadvantages:** thorium-232 is not fissile so must be bred into uranium-233 (slow breeding rate) – U-233 is an alpha emitter so very hazardous to handle – less developed technology with high development costs **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 192-194; AQA spec 3.3.4.1.	

Do not write outside the box

Question	Marking guidance	Marks
06.1	**1 mark each** 1 mark for each of the 2 points below. **1 Fission:** splitting the nuclei of large atoms (U-235, Pu-239) when hit by neutrons **1 Fusion:** joining the nuclei of small atoms (deuterium and tritium - isotopes of hydrogen) to form helium, releasing a neutron and energy - the energy source of stars **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. **Do not credit** • 'nuclear power releases no carbon dioxide' - mining, fuel processing, construction and decommissioning all do	2
06.2	**1 mark each** 1 mark for each of the 2 points below. **1** Deuterium (hydrogen-2) extracted from (sea)water **1** Tritium (hydrogen-3) produced by neutron bombardment of lithium (a lithium blanket around the reactor) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
06.3	**1 mark each** 1 mark for each of the 3 points below. **1** Very high temperature turns hydrogen into a plasma (electrons stripped away) and gives nuclei enough kinetic energy to overcome the repulsion between positive nuclei and collide **1** Vacuum so the plasma is not cooled by air **1** Magnetic field holds the plasma centrally so it does not touch the container walls and cool **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
06.4	**1 mark each** 1 mark for each of the 3 points below. **1** Fuel (deuterium / lithium) is abundant / effectively unlimited **1** Much less radioactive waste (no long-lived fission products) – no chain reaction / meltdown risk – no weapons material **1** Not yet achieved sustained net energy output: reactors (JET) use more energy than they release – ITER aims for 500 MW out from 50 MW in – laser fusion also experimental – commercial viability decades away **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3

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Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 195-196; AQA spec 3.3.4.1.	
07.1	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Fossil fuel technologies:** secondary / tertiary recovery raising recovery from 20% to 60% – directional drilling and subsea wells – ROVs – fracking for tight oil and gas – tar sands / oil shales – coal gasification / liquefaction – enhanced gas recovery – methane hydrates – CCS (pre / post combustion, storage) • **Limits:** still finite (discoveries declining since the 1980s) – rising energy and financial cost of unconventional sources – environmental impacts (aquifers, water use, tremors, land take, habitat damage) – CO2 and other pollution – CCS unproven at scale and impractical for vehicles • **Nuclear:** new uranium extraction (polymer adsorption from seawater, phosphate mining, coal ash) – molten salt reactors – plutonium breeder and thorium reactors increasing fuel use many-fold – fusion as a long-term prospect • **Limits:** cost overruns, decommissioning, waste, public opposition, proliferation, embodied energy of low-grade uranium – fusion not commercial • **Sustainability criteria:** maintaining supply without unacceptable environmental, economic or social impact – comparison with renewables and energy conservation – energy gap argument • **Judgement:** technologies can extend supplies and reduce some impacts but cannot make non-renewables sustainable indefinitely	9

Do not write outside the box

Question	Marking guidance	Marks
	– role as transition / base-load alongside renewables **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content. • **fracking** → pumping fluid into rock at high pressure to release gas or oil • **CCS** → capturing carbon dioxide from flue gases and storing it underground • **adsorption** → nutrient ions sticking to the surfaces of clay and humus **Do not credit** • 'nuclear power releases no carbon dioxide' - mining, fuel processing, construction and decommissioning all do • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts	
	Model answer - Level 3 (8 marks)	
	Technology has repeatedly extended fossil-fuel supplies. Secondary and tertiary recovery raise the proportion of oil recovered from about 20% to 60%; directional drilling, subsea wells and ROVs open deep-water fields; fracking releases tight oil and gas; tar sands and oil shales, coal gasification and liquefaction convert lower-grade resources; and carbon capture and storage could reduce the CO_2 impact. Nuclear fission can be stretched even further: uranium can be adsorbed from seawater with polymers or recovered from phosphate mining and coal ash, molten-salt reactors run more safely, and plutonium breeder and thorium reactors multiply the energy obtainable from each tonne of fuel, with fusion a long-term prospect.	
	None of this makes the resources renewable. Discoveries of conventional oil have declined since the 1980s, unconventional sources cost more energy and money per unit and damage aquifers, use huge volumes of water, cause tremors and take land, and CCS is unproven at scale. Nuclear projects overrun their budgets, decommissioning and long-lived waste remain unsolved, public opposition and proliferation constrain expansion, low-grade uranium has a high embodied energy, and fusion is not commercial.	
	Sustainability means maintaining supply without unacceptable environmental, economic or social impact. New technology can extend supplies for decades and cut some impacts, which matters for closing the energy gap while renewables and conservation scale up, but it cannot make non-renewable resources sustainable indefinitely. Their proper role is as a transition and base-load source alongside renewables, not a permanent solution.	
	Weaker answer - Level 2 (5 marks)	
	New technologies mean fossil fuels can last longer. Fracking gets gas out of shale rock and directional drilling reaches oil under the sea. Carbon capture and storage can stop the carbon dioxide going into the atmosphere. For nuclear power, breeder reactors and thorium reactors can use fuel more efficiently and fusion might work in the future. However fossil fuels are still non-renewable so they will run out eventually, and fracking can pollute groundwater. Nuclear waste is still a problem. So new technologies help but they cannot make fossil fuels and nuclear power fully sustainable.	
	What makes the difference	

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Question	Marking guidance	Marks
	• Level 2 reaches a sensible conclusion but supports it thinly; Level 3 gives specific evidence (20% to 60% recovery, declining discoveries, embodied energy of low-grade uranium).	
	• Level 3 sets out a criterion for sustainability and judges against it - the hallmark of 'evaluate'.	
	• Level 3 places the technologies in context (transition and base load alongside renewables) rather than treating them in isolation.	
07.2	1 mark for the point below. **1** 80-90% **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 183-196; AQA spec 3.3.3-3.3.4.1.	

Do not write outside the box